Dengue Vector SDM Workshop: Setup Guide

Dengue Vector SDM Workshop: Setup Guide

Welcome! In this workshop we will use R to build a first species distribution model for the dengue vector *Aedes aegypti* in Brazil.

The workshop is beginner-friendly, but it does rely on several R packages. Please try to complete this setup before the session. If something fails, do not worry: we will have cached data available during the workshop.

1. Install R

Download and install the latest version of R from:

<https://cran.r-project.org/>

Choose the version for your operating system:

- Windows: "Download R for Windows"
- macOS: "Download R for macOS"

2. Install RStudio Desktop

Download and install RStudio Desktop from:

<https://posit.co/download/rstudio-desktop/>

R is the programming language. RStudio is the interface we will use to write and run R code.

3. Download The Workshop Folder

You should have a folder called:

```
SDM_dengue_Brazil_workshop
```

Put it somewhere easy to find, such as your Desktop or Documents folder.

4. Open The RStudio Project

Inside the workshop folder, double-click:

```
SDM_dengue_Brazil_workshop.Rproj
```

This opens RStudio in the correct folder. This means you should not need to use `setwd()`.

5. Install The R Packages

In RStudio, open:

```
scripts/00_install_packages.R
```

Run the whole script.

The most important packages are:

```
install.packages(c(
  "biomod2",
  "gbm",
  "dplyr",
  "ggplot2",
  "readr",
  "rgbif",
  "sf",
  "terra",
  "viridis",
  "rnaturalearth",
  "rnaturalearthdata",
  "randomForest"
))
```

Optional packages:

```
install.packages(c("CoordinateCleaner", "geodata"))
```

The workshop scripts do not require these optional packages, but they are useful for future SDM work.

6. Check Your Setup

Open and run:

```
scripts/01_setup_check.R
```

If everything is working, you should see a message saying that the setup check passed.

7. Common Installation Problems

Windows

If package installation asks whether to compile from source, choose "No" unless you already have RTools installed.

macOS

Some spatial packages may ask for command line tools. If prompted, install them and try again.

Slow Internet

The workshop includes cached occurrence data and environmental layers. If the live GBIF download fails, you can still complete the model.

8. During The Workshop

The participant-facing lesson is:

```
lesson.html
```

The short intro slide deck is:

```
slides/intro_sdm_dengue_brazil.pptx
```

During the live workshop, we will mostly use one script and run it section by section:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

The other scripts are setup checks, cached-data builders, or optional exports.

Outputs will be saved into:

```
outputs/
```

The model is a teaching model. It estimates environmental suitability for *Aedes aegypti*, not dengue transmission or human disease risk.

Mapping Dengue Vector Suitability In Brazil

```
knitr::opts_chunk$set(
  eval = FALSE,
  echo = TRUE,
  message = FALSE,
  warning = FALSE
)
```

Welcome

In this workshop, we will build a first species distribution model for *Aedes aegypti*, one of the major mosquito vectors associated with dengue transmission.

We will focus on Brazil because it gives us a real public health context, a national-scale map, and enough occurrence data for a meaningful teaching example.

By the end, you should be able to:

- open an RStudio project;
- download occurrence records from GBIF;
- clean and inspect coordinate data;
- load environmental predictors;
- run GLM, GBM, and Random Forest models using `biomod2`;
- build simple ensemble models;
- map predicted environmental suitability;
- explain why suitability is not the same as dengue risk.

The workshop is designed for people who are relatively new to R. Please run code slowly, one section at a time.

What Are We Modeling?

Species distribution models, often called SDMs, estimate where environmental conditions are suitable for a species.

For dengue, this can be useful because the spatial distribution of vectors such as *Aedes aegypti* helps shape where transmission could occur. But this is only one piece of the puzzle.

An SDM for *Aedes aegypti* does **not** directly model:

- human dengue cases;
- mosquito abundance;
- human movement;
- immunity;
- vector control;
- health system reporting.

It estimates environmental suitability for the vector based on the data and predictors we provide.

Open The Project

Open the file:

```
SDM_dengue_Brazil_workshop.Rproj
```

This tells RStudio that the workshop folder is the working directory. You should not need to use `setwd()`.

Run:

```
getwd()
list.files()
```

You should see folders such as `scripts`, `data`, and `outputs`.

Check Setup

Open and run:

```
scripts/01_setup_check.R
```

Or run:

```
source("scripts/01_setup_check.R")
```

If any package is missing, run:

```
source("scripts/00_install_packages.R")
```

Download Occurrence Data From GBIF

GBIF is the Global Biodiversity Information Facility. It provides public biodiversity occurrence records from museums, surveys, citizen science, and other sources.

Today, our target is:

```
target_species <- "Aedes aegypti"
target_country <- "BR"
```

Open the main workshop script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 1** and **Section 2**, then run them line by line.

The important idea is:

```
gbif_count <- rgbif::occ_count(
  scientificName = target_species,
  country = target_country,
  hasCoordinate = TRUE
)
```

Then we download the records:

```
gbif_download <- rgbif::occ_search(
  scientificName = target_species,
  country = target_country,
  hasCoordinate = TRUE,
  limit = min(gbif_count, 10000)
)

occurrences_raw <- gbif_download$data
```

If GBIF Is Slow

Use the cached backup by keeping:

```
download_from_gbif <- FALSE
```

The backup file is:

```
data/occurrences/aedes_aegypti_brazil_clean_backup.csv
```

Clean Occurrence Records

GBIF data are powerful, but they are not perfect. Records can have missing coordinates, duplicated points, or coordinates outside the study region.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 3: Clean and map occurrence records**.

The cleaning steps remove:

- missing coordinates;
- duplicate coordinates;
- coordinates outside Brazil's broad bounding box;
- zero-zero coordinates.

We also keep a workshop-sized sample so the model runs quickly.

```
max_points <- 1000
```

This is not a magic number. It is a teaching choice.

Pause And Think

Look at the occurrence map.

Ask yourself:

- Are records evenly spread across Brazil?
- Are there clusters near cities or research centers?
- Could this reflect sampling effort rather than mosquito ecology?

Load Environmental Predictors

The model needs environmental data for both presence locations and the wider study area.

For the live workshop, we use cached WorldClim bioclimatic predictors cropped to Brazil.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 4: Load environmental predictors**.

The predictors are:

- annual mean temperature;
- temperature seasonality;
- annual precipitation;
- precipitation seasonality.

These predictors are not the only possible choices. They are a simple, teachable starting point.

Run First SDMs With biomod2

Now we fit the models.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 6: Prepare data for biomod2** and continue through the modeling sections slowly.

The model needs:

```
species_xy <- as.matrix(occurrences[, c("decimalLongitude", "decimalLatitude")])
species_presence <- rep(1, nrow(occurrences))
```

Because we only have presence records, `biomod2` creates pseudo-absence points.

```
formatted_data <- BIOMOD_FormatingData(
  resp.name = "Aedes_aegypti",
  resp.var = species_presence,
  resp.xy = species_xy,
  expl.var = predictors,
  PA.nb.rep = 1,
  PA.nb.absences = 1000,
  PA.strategy = "random",
  filter.raster = TRUE,
  seed.val = 42
)
```

We run three example algorithms:

- GLM: Generalized Linear Model
- GBM: Generalized Boosting Model, also called boosted regression trees

- RF: Random Forest

For the live workshop, we keep the models on biomod2's `bigboss` default settings. The script has an optional final section showing where model settings can be changed later.

```
models_to_run <- c("GLM", "GBM", "RF")

model_out <- BIOMOD_Modeling(
  bm.format = formatted_data,
  modeling.id = "first_sdm",
  models = models_to_run,
  CV.strategy = "random",
  CV.nb.rep = 1,
  CV.perc = 0.8,
  OPT.strategy = "bigboss",
  metric.eval = c("TSS", "AUCroc"),
  var.import = 1,
  nb.cpu = 1,
  seed.val = 42
)
```

What Are Pseudo-Absences?

Occurrence data tell us where the species was recorded. They usually do not tell us where the species was truly absent.

Pseudo-absences are background locations used to help the model compare known presence sites with available environmental conditions.

This is useful, but it is also a modeling choice. Different pseudo-absence strategies can change the result.

Build Ensemble Models

Ensemble models combine predictions from the individual models.

In the workshop script, we build two simple ensemble examples:

- `EMmean`: average prediction across selected models
- `EMwmean`: weighted average, where better-performing models get more weight

The key function is:

```
ensemble_out <- BIOMOD_EnsembleModeling(
  bm.mod = model_out,
  models.chosen = "all",
  em.by = "all",
  em.algo = c("EMmean", "EMwmean"),
  metric.select = "all",
```

```
  metric.eval = c("TSS", "AUCroc"),
  var.import = 1
)
```

Project Suitability Across Brazil

After fitting the individual and ensemble models, we predict suitability across the whole study area:

```
projection_out <- BIOMOD_Projection(
  bm.mod = model_out,
  proj.name = "Brazil_current",
  new.env = predictors,
  models.chosen = "all"
)

ensemble_projection_out <- BIOMOD_EnsembleForecasting(
  bm.em = ensemble_out,
  bm.proj = projection_out,
  models.chosen = "all"
)
```

The outputs are saved as:

```
outputs/maps/aedes_aegypti_brazil_suitability.tif
outputs/maps/aedes_aegypti_brazil_ensemble_suitability.tif
```

Map And Interpret

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 9: Project models across Brazil**, **Section 10: Map predictions**, and **Section 11: Interpretation**.

This creates:

```
outputs/maps/aedes_aegypti_brazil_suitability.png
outputs/maps/aedes_aegypti_brazil_ensemble_suitability.png
```

Interpretation Questions

Discuss:

- Where does the model predict high suitability?
- Does that match what you know about *Aedes aegypti*?

- Where might the map be influenced by sampling bias?
- What does the model miss?
- What would you need before using this in a public health decision?

Adapting The Workflow

To model another species, change:

```
target_species <- "Aedes albopictus"
target_country <- "BR"
```

Or choose another country code:

```
target_country <- "ZA"
```

You would also need environmental predictors that cover the new study area.

What To Remember

An SDM is a useful way to connect occurrence data and environmental predictors.

For infectious disease work, SDMs are most powerful when combined with:

- vector biology;
- surveillance data;
- human population data;
- uncertainty checks;
- local epidemiological knowledge.

The map is a starting point for questions, not the final answer.

Data Sources And Provenance

Data Sources And Provenance

This workshop uses public data for teaching species distribution modeling.

Occurrence Data

Species:

```
Aedes aegypti
```

Country:

```
Brazil, GBIF country code BR
```

Source:

```
GBIF occurrence records downloaded through the rgbif R package
```

The facilitator cache was created with:

```
scripts/98_create_cached_workshop_data.R
```

Cached files:

```
data/occurrences/aedes_aegypti_brazil_gbif_raw.csv
data/occurrences/aedes_aegypti_brazil_clean_backup.csv
```

The cached clean backup contains a teaching-sized sample of cleaned coordinate records.

Environmental Predictors

Source:

```
WorldClim v2.1 bioclimatic variables, 10 minute resolution
```

URL used by the cache builder:

```
https://geodata.ucdavis.edu/climate/worldclim/2_1/base/wc2.1_10m_bio.zip
```

Variables used:

```
bio01 annual mean temperature
bio04 temperature seasonality
bio12 annual precipitation
bio15 precipitation seasonality
```

Cached file:

```
data/environment/brazil_bioclim_predictors.tif
```

Boundary Data

Source:

```
Natural Earth country boundary, accessed through rnaturalearth
```

Cached file:

```
data/boundaries/brazil_boundary.gpkg
```

Important Interpretation Note

The model estimates environmental suitability for *Aedes aegypti*. It does not estimate dengue incidence, dengue transmission, mosquito abundance, or future outbreak risk.

Troubleshooting Package Installation

Troubleshooting Package Installation

First Things To Try

Restart RStudio, then run:

```
source("scripts/00_install_packages.R")
source("scripts/01_setup_check.R")
```

Make sure you opened the project file:

```
SDM_dengue_Brazil_workshop.Rproj
```

If `terra` Or `sf` Fails

These packages connect R to geospatial libraries. They usually install cleanly from CRAN binaries, but they can be fussy on older systems.

Try:

```
install.packages("terra", type = "binary")
install.packages("sf", type = "binary")
```

If R says binary packages are not available, install the latest R version from CRAN and try again.

If `biomod2` Fails

Try installing the core modeling packages first:

```
install.packages(c("randomForest", "gbm", "maxnet"))
install.packages("biomod2")
```

The live workshop uses GLM by default, which has the lightest dependency burden.

If GBIF Download Fails

That is okay. Use the cleaned backup data:

```
use_cached_backup <- TRUE
```

in:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

If R Cannot Find Files

Check the top-right of RStudio. The project name should be:

```
SDM_dengue_Brazil_workshop
```

Then run:

```
getwd()
list.files()
```

You should see folders called `scripts`, `data`, and `outputs`.